

Practical -03

Code:

```
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GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork Failed\n");
        return 1;
    }

    if (pid == 0)
    {
        printf("==== CHILD PROCESS =====\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
        printf("Child State : Running\n");

        sleep(1);

        printf("Child State : Terminated\n");
        exit(0);
    }
    else
    {

```

```

    }
    else
    {
        printf("==== PARENT PROCESS =====\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
        printf("Parent State : Waiting\n");

        wait(NULL);

        printf("Child Finished.\n");
        printf("Parent State : Running\n");
        printf("Parent Terminated.\n");
    }

    return 0;
}

```

Output:

```

bash: ./: is a directory
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-03$ ./process_states
==== PARENT PROCESS =====
Parent PID : 1277
Child PID : 1278
Parent State : Waiting
==== CHILD PROCESS =====
Child PID : 1278
Parent PID : 1277
Child State : Running
Child State : Terminated
Child Finished.
Parent State : Running
Parent Terminated.

```

Using ps:

```

bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-03$ ps -ef | grep process_states
bharath+  1282    312  0 03:45 pts/0    00:00:00 grep --color=auto process_states
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-03$ |

```

Using top:

```
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-03$ top
top - 03:46:21 up 50 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7756.6 total, 7141.7 free, 516.7 used, 255.0 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7239.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1285	bharath+	20	0	9280	5728	3512	R	0.5	0.1	0:00.01	top
1	root	20	0	21800	13208	9796	S	0.0	0.2	0:00.72	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3180	2040	1940	S	0.0	0.0	0:00.00	init
39	root	19	-1	66828	15972	14784	S	0.0	0.2	0:00.43	systemd-journal
89	root	20	0	25284	6660	5176	S	0.0	0.1	0:00.25	systemd-udevd
155	systemd+	20	0	21344	13276	11152	S	0.0	0.2	0:00.09	systemd-resolve
156	systemd+	20	0	91036	7920	6956	S	0.0	0.1	0:00.15	systemd-timesyn
165	root	20	0	4244	2680	2432	S	0.0	0.0	0:00.01	cron
166	message+	20	0	9600	5396	4692	S	0.0	0.1	0:00.15	dbus-daemon
180	root	20	0	17980	8884	7844	S	0.0	0.1	0:00.10	systemd-logind
190	syslog	20	0	222516	5948	4552	S	0.0	0.1	0:00.09	rsyslogd
212	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
213	root	20	0	112820	25280	15328	S	0.0	0.3	0:00.10	unattended-upgr
310	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
311	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.18	Relay(312)
312	bharath+	20	0	6728	6056	3704	S	0.0	0.1	0:00.20	bash
313	root	20	0	6700	4656	3876	S	0.0	0.1	0:00.01	login
357	bharath+	20	0	20320	11516	9388	S	0.0	0.1	0:00.10	systemd
358	bharath+	20	0	21168	3596	1868	S	0.0	0.0	0:00.00	(sd-pam)
384	bharath+	20	0	6080	5332	3668	S	0.0	0.1	0:00.01	bash
675	polkitd	20	0	308172	8188	7296	S	0.0	0.1	0:00.10	polkitd

Using proc:

```
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-03$ ls /proc
1      180  39      config.gz      execdomains    kcore          locks          partitions    sys           vmstat
1305   190  6        consoles      fb             key-users      meminfo       pressure     sysrq-trigger zoneinfo
1307   2    675      cpuinfo       filesystems    keys           misc          schedstat    sysvipc
1309   212  89       crypto        fs            kmsg          modules       scsi         thread-self
1334   213  acpi     devices       interrupts     kpagecgroup   mounts        self         timer_list
155    313  buddyinfo diskstats     iomem         kpagecount    mpt          slabinfo     tty
156    357  bus      dma           ioports       kpageflags    mtrr         softirqs     uptime
165    358  cgroups  driver        irq           latency_stats net           stat         version
166    384  cmdline  dynamic_debug kallsyms      loadavg       pagetypeinfo swaps        vmallocinfo
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-03$ |
```